

FEATURE EVALUATION AND SELECTION SUMMARY

1. Introduction

Feature evaluation and selection were conducted to determine which variables should be retained in the final modelling dataset and which variables should be removed. The evaluation considered the findings from exploratory data analysis, correlation analysis, statistical testing, feature engineering, business relevance, interpretability and potential redundancy. The objective was not simply to retain statistically significant variables, but to develop a balanced feature set that provides meaningful information for predictive modelling while avoiding unnecessary duplication.

2. Feature evaluation criteria

Features were evaluated using the following criteria:

- Analytical relevance to the loan prediction problem,
- Relationship with the target variable,
- Statistical evidence from hypothesis testing,
- Correlation with other variable,
- Potential redundancy,
- Interpretability,
- Business relevance,
- Suitability for machine-learning modelling.

The feature was therefore not removed solely because its individual statistical test was not significant. Similarly, a feature was not automatically retained solely because it showed statistical significance.

3. Original numerical features

The following numerical variables were retained:

Feature	Evaluation	Decision
ApplicantIncome	Important applicant financial information, positively related to LoanAmount	Retained
CoapplicantIncome	Provides additional household income information	Retained
LoanAmount	Directly represents the requested loan amount	Retained
Loan_Amount_Term	Represents the requested repayment period	Retained

Credit_History	Important credit-related applicant information and strong candidate predictor	Retained
TotalIncome	Meaningful engineered measure of combined income	Retained
DebtToIncomeRatio	Provides a measure of loan burden relative to income	Retained

These variables provide important financial and credit related information relevant to loan assessment.

4. Original categorical features

The following applicant and application characteristics were retained:

Feature	Statistical/Analytical Finding	Decision
Gender	Not statistically significant in the chi-square analysis	Retained
Married	Statistically significant association with Loan_Status	Retained
Dependents	Not statistically significant	Retained
Education	Statistically significant association with Loan_Status	Retained
Self_Employed	Not statistically significant	Retained
Property_Area	Statistically significant association with Loan_Status	Retained

Although some categorical variables did not demonstrate statistical significance, they were retained because statistical significance alone was not used as the sole feature-selection criterion. These variables may still provide predictive information when combined with other features during machine-learning modelling.

5. Engineered features

Four main engineered features were retained:

TotalIncome

Retained because it combines applicant and coapplicant income and provides a broader representation of financial resources.

IncomeBand

Retained because it provides an interpretable categorical representation of income.

DebtToIncomeRatio

Retained because it provides additional information about the relationship between loan requirements and income.

FamilySizeGroup

Retained because it provides a group representation of family size.

6. Features removed

Two variables were excluded from the final modelling dataset.

6.1 Loan_ID

Loan_ID was removed because it is an identifier rather than an applicant or application characteristic. An identifier does not provide meaningful information about the financial circumstances of an applicant and could introduce noise into a predictive model.

Decision: Removed.

6.2 CreditRiskCategory

CreditRiskCategory was initially considered because it demonstrated a strong statistical association with Loan_Status.

The chi-square test produced:

- $\chi^2 = 180.0665$
- $p < 0.001$

However, CreditRiskCategory was derived from Credit_History. Therefore, retaining both CreditRiskCategory and Credit_History would introduce redundant information into the model. Credit_History was retained as the original predictor, while CreditRiskCategory was removed.

Decision: Removed.

7. Correlation considerations

Correlation analysis was also considered during feature evaluation.

The strongest observed relationships included:

- ApplicantIncome and TotalIncome: $r = 0.8930$,
- LoanAmount and TotalIncome: $r = 0.6246$,
- ApplicantIncome and LoanAmount: $r = 0.5709$.

The strong relationship between ApplicantIncome and TotalIncome was considered during feature refinement because TotalIncome incorporates applicant income. However, TotalIncome was retained because it provides a meaningful combined-income representation that differs conceptually from ApplicantIncome alone. The final decision should therefore be evaluated further during Week 4 modelling using model performance and feature-importance techniques.

8. Target variable

The target variable is:

Loan_Status

It was encoded as:

- 1 = Approved
- 0 = Rejected

The final target distribution is:

Loan Status	Count
Approved	422
Rejected	192

The target variable was not included among the predictors.

9. Final feature set

After feature evaluation and selection, the final dataset contains 22 predictor variables and one target variable.

The predictor variables are:

1. ApplicantIncome
2. CoapplicantIncome
3. LoanAmount
4. Loan_Amount_Term
5. Credit_History
6. TotalIncome
7. DebtToIncomeRatio
8. Gender_Male
9. Married_Yes
10. Dependents_1
11. Dependents_2
12. Dependents_3+
13. Education_Not Graduate
14. Self_Employed_Yes

15. Property_Area_Semiurban
16. Property_Area_Urban
17. IncomeBand_Lower-Middle
18. IncomeBand_Upper-Middle
19. IncomeBand_High
20. FamilySizeGroup_Medium Family
21. FamilySizeGroup_Small Family
22. FamilySizeGroup_Unknown

The target variable is:

Loan_Status

10. Final dataset validation

Following feature selection, categorical variables were encoded and numerical variables were scaled where applicable.

The final validation produced:

- Dataset shape: 614 × 23
- Predictor variables: 22
- Target variable: 1
- Total missing values: 0
- Loan_Status missing values: 0

The final dataset therefore satisfies the basic structural requirements for machine-learning modelling.

11. Selection outcome

The feature evaluation process resulted in a refined feature set that balances statistical evidence, business relevance, interpretability and avoidance of redundant information.

The main feature selection decisions were:

- Retain meaningful financial variables,
- Retain Credit_History as the original credit-related predictor,
- Retain statistically significant categorical variables such as Married, Education and Property_Area,

- Retain engineered variables that provide meaningful additional representations,
- Remove Loan_ID because it is an identifier,
- Remove CreditRiskCategory because it duplicates Credit_History,
- Encode categorical predictors.
- Scale numerical predictors where applicable.

12. Conclusion

The feature evaluation and selection process provided a structured approach to refining the loan prediction dataset. Rather than relying on a single statistical criterion, the selection process considered the combined evidence from exploratory analysis, statistical analysis, correlations, feature engineering and business relevance. The resulting dataset contains 614 observations and 23 variables, including 22 predictors and the encoded Loan_Status target. This refined dataset will serve as the foundation for machine-learning model development and feature importance evaluation in week 4.